

RCR Checklist

1. DSEVAL

ID	Quantity required	requirement
REQ01	5	Train data path: '{data_dir}/DSEval/datasets/01_bank_customer_churn/train.csv' test data path: '{data_dir}/DSEval/datasets/01_bank_customer_churn/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out: 1. Drop the columns that will not be interesting to models, such as "CustomerId", "Complain", and "Surname". 2. Use one-hot encoding for "Geography" and "Card Type". 3. Construct a model using XGBoost to predict categorical target variable "Exited". 4. Make predictions on the test set. 5. Compute area under ROC curve.
REQ02	6	Train data path: '{data_dir}/DSEval/datasets/02_cardiovascular_diseases/train.csv' test data path: '{data_dir}/DSEval/datasets/02_cardiovascular_diseases/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out: 1. Drop all rows with missing values from the dataset. And then drop the duplicates. 2. Perform feature construction to build a new feature named "BMI_Category" to categorize the BMI (Body Mass Index) of each individual into one of four categories: "Underweight", "Normal weight", "Overweight", and "Obesity". This is based on the following ranges: - "Underweight" is a BMI less than 18.5, - "Normal weight" is a BMI from 18.5 to 24.9, - "Overweight" is a BMI from 25 to 29.9, - "Obesity" is a BMI of 30 or higher. Use category dtype for "BMI_Category". 3. Try to predict Heart_Disease based on all other variables.

		Construct a model to predict Heart_Disease. Fit the model on the training set. Make predictions on the test set. Compute area under ROC curve.
REQ0 3	6	Train data path: '{data_dir}/DSEval/datasets/03_coffee_quality/train.csv' test data path: '{data_dir}/DSEval/datasets/03_coffee_quality/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out: Drop duplicated rows from the dataset. Drop the following columns from the dataset: "ID", "ICO Number", "Owner", "Region", "Certification Contact", "Certification Address", "Farm Name", "Lot Number", "Mill", "ICO Number", "Producer", "Company", "Expiration", "Harvest Year", "Unnamed: 0", "Number of Bags", "Bag Weight", "In-Country Partner", "Country of Origin", "Processing Method", "Grading Date", "Variety", "Status", "Defects", "Uniformity", "Clean Cup", "Sweetness", and "Certification Body". - For categorical column "Color", use one-hot encoding. - Try to predict Total Cup Points. Build a regressor on the training set with RandomForestRegressor. Make predictions on the test set. Evaluate the mean-squared error for the model on the testing set.
REQ0 4	7	Train data path: '{data_dir}/DSEval/datasets/04_diabetes/train.csv' test data path: '{data_dir}/DSEval/datasets/04_diabetes/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out: Drop duplicate rows from the dataset. - Perform one-hot encoding on the "gender" and "smoking_history" columns. Rank the most important features. Build a random forest classifier to predict diabetes using all features. Fit the model on the training set. Make predictions on the test set. Compute area under ROC curve. Show the confusion matrix.
REQ0 5		Train data path: '{data_dir}/DSEval/datasets/05_patient_profile/train.csv' test data path: '{data_dir}/DSEval/datasets/05_patient_profile/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out:

	5	you need build two logistic regression models to predict categorical target variable 'outcome variable'. 1. categorical features (except "Disease" column) into numerical features using one-hot encoding. 2. Use a feature selection technique to select the most important 5 features. 3. Build a logistic regression model1 using all features one-hot encoded. 4. Build a logistic regression model2 using only the most important 5 features. 5. Evaluate the performance of the model1 and model2 on the test set.
REQ0 6	6	Train data path: '{data_dir}/DSEval/datasets/06_employee/train.csv' test data path: '{data_dir}/DSEval/datasets/06_employee/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out: 1. Perform feature construction to build a new feature named "YearsInCompany" to represent the number of years each employee has been in the company. It is calculated as 2023 minus the 'JoiningYear'. 2. Encode the categorical columns using one-hot encoding. 3. Build a random forest classifier to predict the 'LeaveOrNot' variable. 4. Make predictions on the test set. 5. Compute area under ROC curve. 6. Confusion matrix showing predicted results.
REQ0 7	4	Train data path: '{data_dir}/DSEval/datasets/07_sustainable_energy/train.csv' test data path: '{data_dir}/DSEval/datasets/07_sustainable_energy/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out: 1. The features are all other columns except 'Entity' and 'Year'. 2. The target variable is 'Renewable energy share in the total final energy consumption (%)'. Build a linear regression model to predict the outcome variable using all features. Fit the model on the training set. 3. Make predictions on the test set. 4. Evaluate the performance of the linear regression model on the test set using metrics including RMSE.
REQ0 8	4	Train data path: '{data_dir}/DSEval/datasets/08_heart_attack/train.csv' test data path: '{data_dir}/DSEval/datasets/08_heart_attack/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out:

		1. Convert the values in the data for the category column 'Diet'. Use 1, 2, 3 for 'Healthy', 'Average', and 'Unhealthy' respectively. 2. Build a RandomForest model to predict the Heart Attack Risk using all features. Fit the model on the training set. 3. Make predictions on the test set. 4. Evaluate the performance of the RandomForest model on the test set. Return roc_auc scores in a tuple.
REQ09	3	Train data path: '{data_dir}/DSEval/datasets/09_housing_price/train.csv' test data path: '{data_dir}/DSEval/datasets/09_housing_price/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out: 1. loading 'housing_price_prediction_train' data and loading 'housing_price_prediction_test' data 2. Build a linear regression model to predict the housing price. Fit the model on the training set. 3. Make predictions on the test set. 4. Evaluate the performance of the model on the test set using RMSE.
REQ10	4	Train data path: '{data_dir}/DSEval/datasets/10_life_expectancy/train.csv' test data path: '{data_dir}/DSEval/datasets/10_life_expectancy/test.csv'. Use the training data and test data to go through all of the following tasks, taking care not to leave anything out: 1. Perform feature construction to build a new feature named "GDP_bins" to categorize the GDP into several bins. The GDP bins are defined as follows: • "Low GDP": GDP <= 1000 • "Medium GDP": 1000 < GDP <= 10000 • "High GDP": GDP > 10000 2. Create a predictive model to predict life expectancy. Use Linear regression as the model. Fit the model on the training set. 3. Make predictions on the test set. 4. Evaluate the performance of the model using R-squared score.

2. ML_BENCHMARK

ID	Quantity	requirement

	required	
IRIS_REQ01	1	Run data analysis on sklearn Iris dataset, include a plot
WINES_RECOGNITION_REQ02	4	Run data analysis on sklearn Wine recognition dataset, include a plot , and train a model to predict wine class with 20% as test set, and show prediction accuracy
BREAST_CANCER_WISCONSIN_REQ03	4	Run data analysis on sklearn Wisconsin Breast Cancer dataset, include a plot , train a model to predict targets (20% as validation), and show validation accuracy
TITANIC_REQ04	5	This is a titanic passenger survival dataset, your goal is to predict passenger survival outcome. The target column is Survived. Perform data analysis, data preprocessing , feature engineering , and modeling to predict the target. Report accuracy on the eval data. Train data path: '{data_dir}/ml_benchmark/04_titanic/split_train.csv', eval data path: '{data_dir}/ml_benchmark/04_titanic/split_eval.csv'.
HOUSE_PRICES_ADVANCED_REGRESSION_TECHNIQUES_REQ05	5	This is a house price dataset, your goal is to predict the sale price of a property based on its features. The target column is SalePrice. Perform data analysis, data preprocessing , feature engineering , and modeling to predict the target. Report RMSE between the logarithm of the predicted value and the logarithm of the observed sales price on the eval data. Train data path: '{data_dir}/ml_benchmark/05_house_prices/split_train.csv', eval data path: '{data_dir}/ml_benchmark/05_house_prices/split_eval.csv'.
SANTANDER_CUSTOMER_TRANSACTION_PREDICTION_REQ06	5	This is a customers financial dataset. Your goal is to predict which customers will make a specific transaction in the future. The target column is target. Perform data analysis, data preprocessing , feature engineering , and modeling to predict the target. Report AUC on the eval data. Train data path: '{data_dir}/ml_benchmark/06_santander_customer/split_train.csv', eval data path: '{data_dir}/ml_benchmark/06_santander_customer/split_eval.csv'.
ICR_IDENTITY_AGE_RELATED_CONDITIONS_REQ07	5	This is a medical dataset with over fifty anonymized health characteristics linked to three age-related conditions. Your goal is to predict whether a subject has or has not been diagnosed with one of these conditions. The target column is Class. Perform data analysis, data preprocessing , feature engineering , and modeling to predict the target. Report F1 Score on the eval data. Train data path: '{data_dir}/ml_benchmark/07_icr_identify/split_train.csv', eval data path: '{data_dir}/ml_benchmark/07_icr_identify/split_eval.csv'.

SANTANDER_VALUE_PREDICTION_CHALLENGE_REQ08	5	This is a customers financial dataset. Your goal is to predict the value of transactions for each potential customer. The target column is target. Perform data analysis, data preprocessing, feature engineering, and modeling to predict the target. Report RMSLE on the eval data. Train data path: <code>'{data_dir}/ml_benchmark/08_santander_value/split_train.csv'</code>, eval data path: <code>'{data_dir}/ml_benchmark/08_santander_value/split_eval.csv'</code>.
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3. NL2WORKFLOW_CI

ID	Quantity required	requirement
01_bank_customer_churn	5	I need you to use the training data of bank credit loans to build a random forest model that can predict whether customers will churn. Then, use this model to predict the test data, and save the predicted results. Finally, show the confusion matrix of the predicted results. Finally, report the auc_roc of the model on the test set. The training dataset address: <code>r'{data_dir}/NL2Workflow/datasets/tabular/bank_customer_churn/train.csv'</code>.测试集地址: <code>r'{data_dir}/NL2Workflow/datasets/tabular/bank_customer_churn/test.csv'</code>.
02_smoker_status_Bio_Signals	3	Please use the biological signal training dataset to predict whether an individual is a smoker. Your ultimate goal is to establish an XGboost model that can accurately determine the smoking status of individuals. Learning Rate uses 0.1, and the learner uses the DART model ... Finally, report the auc_roc of the model on the test set. training dataset address: <code>r'{data_dir}/NL2Workflow/datasets/tabular/smoker_status_Bio_Signals/train.csv'</code>.测试集地址: <code>r'{data_dir}/NL2Workflow/datasets/tabular/smoker_status_Bio_Signals/test.csv'</code>.
03_kidney_stone	4	You need to use human kidney stone-related indicator data, develop a Machine Learning model to predict whether a person has kidney stones. I need you to add a step of data partitioning data processing before Model Training, and finally use the partitioned validation set to evaluate the model. Finally report the auc_roc of the model on the test set. Training set address: <code>r'{data_dir}/NL2Workflow/datasets/tabular/kidney_stone/train.csv'</code>.测试集地址: <code>r'{data_dir}/NL2Workflow/datasets/tabular/kidney_stone/test.csv'</code>.
04_reservation_cancel	4	Your goal is to help my hotel better predict whether customers will cancel their bookings. You first need to load the training data and test data for hotel bookings,

ancellation		train a LightGBM model using the training data, then use the LightGBM model to make predictions on the test data for hotel bookings, and finally save the prediction results. Finally, report the auc_roc of the model on the test set. 训练集地址: r'{data_dir}/NL2Workflow/datasets/tabular/reservation_cancellation/train.csv'。测试集地址: r'{data_dir}/NL2Workflow/datasets/tabular/reservation_cancellation/test.csv'。
05_obesity_CVD_risk	5	Your goal is to train a model to predict an individual's obesity risk level based on various factors related to obesity, such as diet, lifestyle, etc. During feature engineering, I need you to perform feature construction and feature selection ... Finally, report the accuracy of the model on the test set . Training dataset address: r '{data_dir}/NL2Workflow/datasets/tabular/obesity_CVD_risk/train.csv'.测试集地址: r'{data_dir}/NL2Workflow/datasets/tabular/obesity_CVD_risk/test.csv'。
06_cirrhosis_patient	4	Your goal is to build a Machine Learning model to predict the survival status of patients with cirrhosis. I need you to divide the dataset into training dataset and validation set after processing the dataset, so that the model can be trained with the divided training data and evaluated with the validation data Finally, report the logloss of the model on the test set . The training dataset address: r '{data_dir}/NL2Workflow/datasets/tabular/cirrhosis_patient/train.csv'.测试集地址: r'{data_dir}/NL2Workflow/datasets/tabular/cirrhosis_patient/test.csv'。
07_horse_health_outcomes	4	Predict the health status of horses based on the detection data of various indicators of their health status. I need you to use the training dataset to train the model , then use the model to predict the test data, and finally save the prediction results . Finally report the f1 of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/horse_health_outcomes/train.csv'. 测试集地址: r'{data_dir}/NL2Workflow/datasets/tabular/horse_health_outcomes/test.csv'。
08_monster	4	I need you to use the relevant indicator data of monsters to build a Machine Learning model that can distinguish monster types. Then, use this model to predict the test data, and finally save the prediction results . Finally report the accuracy of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/monster/train.csv'. 测试集地址: r'{data_dir}/NL2Workflow/datasets/tabular/monster/test.csv'。
09_crime_category	4	Your goal is to train a model that can predict crime categories based on nearly 12 years of crime training and test data from all communities in San Francisco. Then, use this model to predict the test dataset, and finally save the prediction results ... Finally, report the logloss of the model on the test set . The training dataset address is: r '{data_dir}/NL2Workflow/datasets/tabular/crime_category/train.csv'.测试集地址: r'{data_dir}/NL2Workflow/datasets/tabular/crime_category/test.csv'。
10_abalone	5	Please build a model to predict the age of abalone based on various physical measurement data. After performing one-hot encoding , divide the dataset , and finally evaluate the trained model using the divided data . Finally report the rmlse of the model on the test set . Training set address:

		r'{data_dir}/NL2Workflow/datasets/tabular/abalone/train.csv'. 测试集地址: r'{data_dir}/NL2Workflow/datasets/tabular/abalone/test.csv'。
11_mohs_hardness	3	Your task is to train a neural network model to predict the Mohs hardness value of minerals based on relevant characteristics of mineral Mohs hardness. I hope your activation function uses the hyperbolic tangent function , and the optimization method uses stochastic gradient descent . Finally report the medae of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/mohs_hardness/train.csv'. 测试集地址: r'{data_dir}/NL2Workflow/datasets/tabular/mohs_hardness/test.csv'。
12_crab_age	5	Your task is to use the training dataset of crab age to build a random forest model to predict the age of crabs. Then, use this model to perform prediction on the test dataset of crab age, and finally save the prediction results . The dataset needs to be one-hot encoded . Finally report the MAE of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/crab_age/train.csv'. 测试集地址: r'{data_dir}/NL2Workflow/datasets/tabular/crab_age/test.csv'。
13_wild_blueberry_yield	6	Your task is to build a Machine Learning model to help me predict the yield of wild blueberries. The processing method of feature engineering is: feature construction and feature selection . After feature engineering processing, the data needs to be sampled , and finally the sampled data is used to evaluate the model Finally, report the mae of the model on the test set . Training dataset address: r '{data_dir}/NL2Workflow/datasets/tabular/wild_blueberry_yield/train.csv'. 测试集地址: r '{data_dir}/NL2Workflow/datasets/tabular/wild_blueberry_yield/test.csv'。
14_media_campaign_cost	4	Your task is to use the training data of media campaign costs to build a Machine Learning model to help us predict the media campaign costs of food supermarkets. Then, use this model to make predictions on the test dataset of media campaign costs, and finally save the prediction results . Finally report the rmlse of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/media_campaign_cost/train.csv'. 测试集地址: r '{data_dir}/NL2Workflow/datasets/tabular/media_campaign_cost/test.csv'。

4. NL2WORKFLOW_UCI

ID	Quantity	requirement
	required	
01_bank_cus	2	Your task is to build a Machine Learning model that can predict whether a customer will churn based on various information about bank customers, such as account activity, service usage, etc. Finally report the auc_roc of the model on the test set .

tomer_churn		Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/bank_customer_churn/train.csv'.
02_smoker_status_Bio_Signals	2	Please use the relevant detection data of biological signals to train a Machine Learning model for predicting whether a person is a smoker. Finally report the auc_roc of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/smoker_status_Bio_Signals/train.csv'.
03_kidney_stone	2	You need to use human kidney stone-related indicator data, develop a Machine Learning model to predict whether a person has kidney stones. Finally, report the auc_roc of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/kidney_stone/train.csv'.
04_reservation_cancellation	2	Your task is to build a Machine Learning model to predict whether a customer will cancel a reservation based on the customer's previous hotel reservation-related data. Finally report the auc_roc of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/reservation_cancellation/train.csv'.
05_obesity_CVD_risk	2	Your goal is to train a model using data on various factors related to obesity, such as diet, lifestyle, etc. , to predict an individual's obesity risk level. Finally, report the accuracy of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/obesity_CVD_risk/train.csv'.
06_cirrhosis_patient	2	Cirrhosis is caused by long-term damage to the liver, leading to extensive scar formation, usually due to diseases such as hepatitis or long-term alcohol consumption. We are conducting a study on primary biliary cirrhosis. We hope you can build a model to predict the survival status of cirrhosis patients using 17 clinical features. Finally report the logloss of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/cirrhosis_patient/train.csv'.
07_horse_health_outcomes	2	Your goal is to train a Machine Learning model to predict the health status of horses based on the detection data of various indicators of their health status. Finally report the model's f1 on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/horse_health_outcomes/train.csv'.
08_monster	2	Your task is to train a model to distinguish whether a monster is a ghoul, ghost, or goblin based on its bone length, decay, soullessness, and other feature data... and finally report the accuracy of the model on the test set . Training dataset address: r' {data_dir}/NL2Workflow/datasets/tabular/monster/train.csv'.
09_crime_category	2	Please train a Machine Learning model to predict crime categories based on crime reports from all communities in San Francisco for the past 12 years... Finally, report the logloss of the model on the test set . Training dataset address: r' {data_dir}/NL2Workflow/datasets/tabular/crime_category/train.csv'.

10_abalone	2	Please build a model to predict the age of abalone based on various physical measurement data. Finally report the rmlse of the model on the test set . Training set address: r'{data_dir}/NL2Workflow/datasets/tabular/abalone/train.csv'.
11_mohs_hardness	2	Your task is to train a Machine Learning model to predict the Mohs hardness of a mineral based on the relevant properties of the mineral Mohs hardness... and finally report the model's medae on the test set . The training dataset address is: r '{data_dir}/NL2Workflow/datasets/tabular/mohs_hardness/train.csv'.
12_crab_age	2	For commercial crab farmers, knowing the appropriate age of crabs helps them decide whether and when to harvest them. After a certain age, the growth of crab physical characteristics is almost negligible, so timing the harvest is very important to reduce costs and increase profits. Now, I need you to help me build a model to predict the age of crabs... and finally report the mae of the model on the test set . Training dataset address: r '{data_dir}/NL2Workflow/datasets/tabular/crab_age/train.csv'.
13_wild_blueberry_yield	2	Your task is to build a Machine Learning model that helps me predict the yield of wild blueberries... and finally report the mae of the model on the test set . Training dataset address: r '{data_dir}/NL2Workflow/datasets/tabular/wild_blueberry_yield/train.csv'.
14_media_campaign_cost	2	In the 1980s, Carden & Cherry used Ernest's image to advertise convenience stores. Your task is to build a Machine Learning model based on the training data of media promotion costs to help us predict the media promotion costs of food stores... Finally report the rmlse of the model on the test set . The training dataset address is: r '{data_dir}/NL2Workflow/datasets/tabular/media_campaign_cost/train.csv'.